

Mahmoud Shoaib Abd Altwab Mohammed

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LinkedIn — GitHub

Career Objective

Computer Science student at the Egyptian Chinese University with hands-on experience in **Computer Vision, Deep Learning, and Machine Learning**. Experienced in developing image classification and video-based object tracking systems using Python, TensorFlow, OpenCV, and Transfer Learning. Strong interest in building practical **Computer Vision and AI solutions** and gaining hands-on industry experience.

Education

Egyptian Chinese University (ECU)

Expected Graduation: 2027

Bachelor of Computer Science and Information Technology

Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Machine Learning

Experience

Flight Delay Prediction using Machine Learning

- Developed an end-to-end machine learning system for predicting flight delays from historical aviation data using multiple classification models.

Tools: Python, Scikit-learn, Pandas, NumPy

Real-Time Movie Recommendation System

- Developed a real-time recommendation platform that processes user interactions and delivers personalized movie recommendations through an event-driven architecture.

Tools: Python, Kafka, Redis, FastAPI, Streamlit, Docker

Courses & Certificates

Artificial Intelligence & Machine learning Diploma

Route IT Training Center, Egypt

Technical Skills

Programming: Python, C++, SQL

Core Concepts: Object-Oriented Programming (OOP), Data Structures and Algorithms

Computer Vision: Image Classification, Image Processing, Video Processing, Object Detection, Object Tracking, OpenCV

Deep Learning: CNNs, Transfer Learning, ResNet50, TensorFlow/Keras, Grad-CAM

Machine Learning: Machine Learning, Scikit-learn, Explainable AI, Recommendation Systems

Frameworks & Tools: Pandas, NumPy, Streamlit, FastAPI, Docker, Git, GitHub

Projects

Brain Tumor Classification & Visualization

GitHub

- Developed a deep learning system for brain MRI image classification using Transfer Learning with ResNet50, including image preprocessing and augmentation.
- Implemented Grad-CAM visualizations to highlight image regions influencing model predictions and built an interactive Streamlit interface for inference and visualization.

Tools: Python, TensorFlow, ResNet50, OpenCV, Grad-CAM, Streamlit

Agentic Customer Intelligence Platform

- Designed an AI-driven customer intelligence platform for analyzing customer behavior and supporting data-driven insights through a modular architecture.

Tools: Python, Scikit-learn, SQLite, SHAP, SARIMA

Object Tracking System

GitHub

- Developed a computer vision system for detecting and tracking moving objects in real-time video streams using OpenCV.
- Implemented background subtraction and contour-based detection techniques for extracting and analyzing object motion across video frames.

Tools: Python, OpenCV, NumPy, Streamlit